Export of SUBMISSION.md (scripts/export_pdf.py); no results recomputed. Source paths refer to the repository.

Beyond Words: Detecting AI-Agent Collusion Through Information Theory and Behavioral Evidence

Dennis Toma — Apart Research AI Collusion Sprint, 23–25 October 2026

Merged submission of two sprint projects: Q1.4 (information-theoretic, prices-only detection) and Q2.2 (auditing what agents say together with what they do). Every number is taken from the frozen source repos at the commits listed in SOURCES.md, which maps each number to its JSON file and key.

Abstract

We put LLM sellers in four-seller repeated pricing markets, offer one pair a private chat channel (never telling them to collude), and ask whether an auditor can tell — from prices, from public messages, or from both. In short:

- **Being offered a private channel shows up as a higher price level.** Markup over the Nash price rose in all 9 of Q1.4's main sets (7 seller models; per-set CIs, uncorrected), and a simple max-pair markup screen had a higher point-estimate AUROC than a conditional-dependence detector in all 9 (0.75–0.99 vs 0.34–0.72; paired difference 95% CI > 0 in 8/9).
- **We detect no contemporaneous co-movement.** The dependence detector (pCMI), which flags and localises every pair in the scripted shared-signal positive control, is at chance in a 192-game seven-factor grid (0.51 [0.43, 0.59]), in pre-registered 60-round games (0.34 [0.18, 0.52]) and on agreements whose prices were kept ("executed", 48/149; 0.42 [0.33, 0.53]). Lagged dependence (transfer entropy) is weakly above chance in the grid but below level (§5.2).
- **A zero-shot LLM auditor shows no resolved gain from public messages.** Adding them to prices changes AUROC by −0.01 [−0.09, +0.06] (locked test, n = 95) and −0.07 [−0.17, +0.02] (fresh replications, post hoc pooling). In Qwen families that invite rivals publicly, the text view is *inverted* (AUROC 0.15–0.21, CI below 0.5). (Text did work on an external negotiation dataset where the detected act happens in the chat; §5.3.)
- **Neither signal is specific to collusion.** An unread private pricing notepad produces about two thirds of the channel's premium over rounds 2–12 (point ratio 0.65, CI [0.34, 1.32]); with memory and profit framing, 19/20 gpt-6-luna markets *without* a channel price ≥ Nash + 2; level screens also flag legitimate cost asymmetry.

Price-level screens could therefore serve as a cheap first-stage audit trigger (not evaluated end to end) whose reference must be matched to the seller model and prompt — not evidence of an agreement.

Glossary. **AUROC**: probability that a random positive example scores above a random negative one, ties counting half; the positive class (usually channel assignment) is stated per analysis (0.5 = chance; < 0.5 = inverted). **Level / max-pair markup**: the highest pair's mean price above the per-round Nash price. **pCMI**: max-pair Gaussian conditional mutual information of prices given public state and all sellers' lagged prices. **TE**: transfer entropy (lagged dependence). **Executed**: the judge found a private agreement *and* the pair's prices then followed it (Q1.4 rule, §3). **Nash + 2**: a price at least 2 above the competitive (Nash) price. **C − A**: channel minus no-channel arm. **E*nn* / P2.*n***: experiment / section IDs in the source repos (SOURCES.md).

1. Introduction

Algorithmic pricing agents built on LLMs reach supracompetitive prices without instruction (Fish et al. 2024) and change behaviour when they can communicate (Agrawal et al. 2025). An overseer will mostly see two things: the agents' *actions* (prices) and what they *say* in public. Two detection strategies follow:

1. **Information theory on actions** (Q1.4). Colluders should share information beyond what public state explains, so their prices should be conditionally dependent (Bonjour et al. 2022). Such a detector is

blind to message content: it does not inspect wording directly (resistance to strategic evasion was not tested).
2. **Say + do auditing** (Q2.2). An LLM auditor reading both public messages and prices should catch inconsistencies between what agents say and what they do, and beat single-stream auditors.

We test both on LLM pricing markets, cross-apply each project's detectors to the other's data, and keep three label levels apart: *assigned opportunity* (channel offered), *verified agreement* (a judge reads the private chat) and *realised coordination* (prices follow the agreement).

Contributions.

- A prices-only conditional-dependence detector (pCMI, with shift and same-model surrogate nulls) that passes a scripted positive control and a hard-negative battery (common shocks, shared policies, legitimate leader–follower pricing), plus a map of what it cannot separate (policy copies sharing a seed).
- Evidence across 9 main LLM sets, a 192-game factorial grid and a pre-registered long-game test that channel assignment shows up as a **price level**, with no detected contemporaneous co-movement.
- A registered say-vs-do comparison showing that public text gives a zero-shot auditor no resolved gain and can be **inverted**.
- Confound analyses (prompt artefacts, placebo, tacit high prices, cost asymmetry, prevalence) showing why a price flag is an audit trigger, not proof.

2. Related work

Dependence as a collusion signal. Bonjour et al. (2022) measure collusion as statistical dependence between agents' actions; our pCMI is a conditioned pairwise version. Banchio & Mantegazza (2023) show supracompetitive play from "spontaneous coupling" without agreement — the case our shared-shock and shared-policy hard negatives target. Tailor (2025) applies mutual-information detectors to steganographic message channels; we apply the same estimator family to actions. Transfer entropy follows Schreiber (2000). *LLM pricing collusion.* Fish et al. (2024), Agrawal et al. (2025; double auctions), Lin et al. (2024; market division expressed in actions, not messages), Dhanda (2026; price caps as focal points). *Shared models.* Cao & Hu (2026) show supracompetitive prices from a shared LLM advisor without coordination, motivating our same-model references. *Definitions.* Rose et al. (2026): communication within a subgroup leading to actions harmful to outsiders; Calvano et al. (2020) and Green & Porter (1984) for repeated-pricing benchmarks. *Message-side collusion.* Motwani et al. (2024) and Mathew et al. (2024) study secret/steganographic collusion in messages; SALT-NLP negotiation data serve as our external text case.

3. Setup

Markets. Two separate four-seller repeated price markets with multinomial-logit demand and integer prices; cost and demand state are generally public, but Q1.4's E70 grid varies cost visibility and rival-price visibility. They are not one shared market; numbers are tagged with their project.

- *Q1.4 market* (`sim/market.py`): cost 6/8, static Nash price 10 (cost 8: 12), joint-profit optimum 16; T = 20 in most LLM sets (40–60 in long-game sets). The public message board is **off** by default; it is on only in the E70 grid and the E59 board arm.
- *Q2.2 market* (`llm_market.py`): 12 rounds (40 in the strong-framing arm), prices 6–30, static Nash ≈ 11, joint monopoly ≈ 19; arms `private` (pair has a channel), `none`, and `legit` (publicly announced cost-saving partnership, hard negative). A public board exists in every arm.

Cross-application: Q1.4 scored Q2.2's `llm_v3` market; Q2.2 audited all 192 games of Q1.4's E70 grid (E113). The "words" results therefore rest on Q2.2's markets plus E70, not on Q1.4's nine main sets.

LLM sellers. Q1.4's nine main sets use seven seller models: gpt-6-luna (main), gpt-5.6-luna, deepseek-v4.1-flash, and base and abliterated (refusal-removed) Qwen3.8-27B and Qwen3.6. Q2.2's main market uses gpt-6-luna and adds family arms (Claude Sonnet, deepseek, Qwen3.8 base/abliterated, gpt-6.1-sol, mimo). Profit-maximisation / economics-lab framing after Fish et al. (2024).

Labels. A gpt-6.1-sol judge reads only the private chat (never prices) → explicit agreement / information exchange / none. Claude Sonnet 5.5 re-judged Q2.2's 115 chats and gave the same binary label in 115/115, but each dataset was single-class, the 3-class κ was only 0.33–0.38, and a second check (E84d) agreed on agreed-yes/no in 25/30. Q1.4's judge labels have no cross-family check.

term	rule	used for
executed (Q1.4)	judge-labelled agreement AND both pair firms ≥ agreed − 1 in ≥ ½ of post-agreement rounds AND second-half pair mean ≥ Nash + 2	48/149 executed games; E70 executed subsets. 27 variants give 13–78/149 (E71)
strict-executed (Q2.2)	registered strict rule (E99-amend): judged agreement with floor F ≥ 13; ≥ 10 later rounds; both firms price ≥ F − 1 in ≥ ½ of those rounds; post-agreement pair mean excess over Nash ≥ 2	13/20 channel pairs (10–16/20 under one-at-a-time threshold changes)
kept / followed (Q2.2)	E84 follow rule on the locked test	14/50 communication-positive runs
tacit-high (Q2.2)	strong-framing no-channel market whose mean price minus cost-adjusted Nash over rounds 11–40 is ≥ 2 (a price threshold, not verified coordination)	19/20 strong-framing markets

Splits and discipline. Q1.4 detector settings were tuned only on scripted dev seeds and evaluated once on locked seeds; no setting was tuned on LLM data. Q2.2's auditors are zero-shot with one prompt per view, evaluated once on a locked test split (runs 0030–0124, n = 95) — locked but not fully blind (four disclosed exposures, including the post-hoc choice of the premium statistic) — and replicated on fresh, pre-registered splits. Most Phase 2 experiments were declared before their data; post-hoc analyses are labelled.

4. Methods

Price detectors (Q1.4, prices only). Per pair: |Pearson|, plug-in MI, CMI given public state, and **pCMI** = $-\frac{1}{2}\log(1-\rho^2)$ of residuals after regressing each seller's price on state dummies and all four lagged prices; transfer entropy (TE, lagged, max over directions). Two nulls: a circular-shift null (B = 200, relative offsets ≥ 3 rounds), and a **same-model surrogate null** ranking the run's max-pair score against no-channel games of the same model and shock family. Level screens: *max-pair markup* over the per-round Nash price and *pair_level* (pair mean minus rivals). An independent reimplementation agreed on 422/424 run × statistic cells to < 1e-9 on the v1 datasets.

Auditors (Q2.2). One zero-shot gpt-6.1-sol auditor, same task text, six views: `text` (public messages

- public facts), `action` (prices, costs, demand, profit), `action_stats` (+ computed pair-statistics table), `combined` (text ∪ action), `combined_stats`, `structured`. Registered primary contrast: Δ_hybrid = AUROC(combined) − max(AUROC(text), AUROC(action)), on communication labels (positive = explicit agreement or information exchange; on the locked test all 50 private runs are judged y = 1 — 48 explicit, 2 information exchange — so judge and channel labels coincide there).

Statistics. Run-level AUROC with class-stratified game bootstrap 95% CIs; Wilson intervals for flag rates; Jeffreys for precision. CIs are per-set and uncorrected for multiplicity unless stated.

5. Results

5.1 On scripted sellers, conditioning works (positive control; Q1.4)

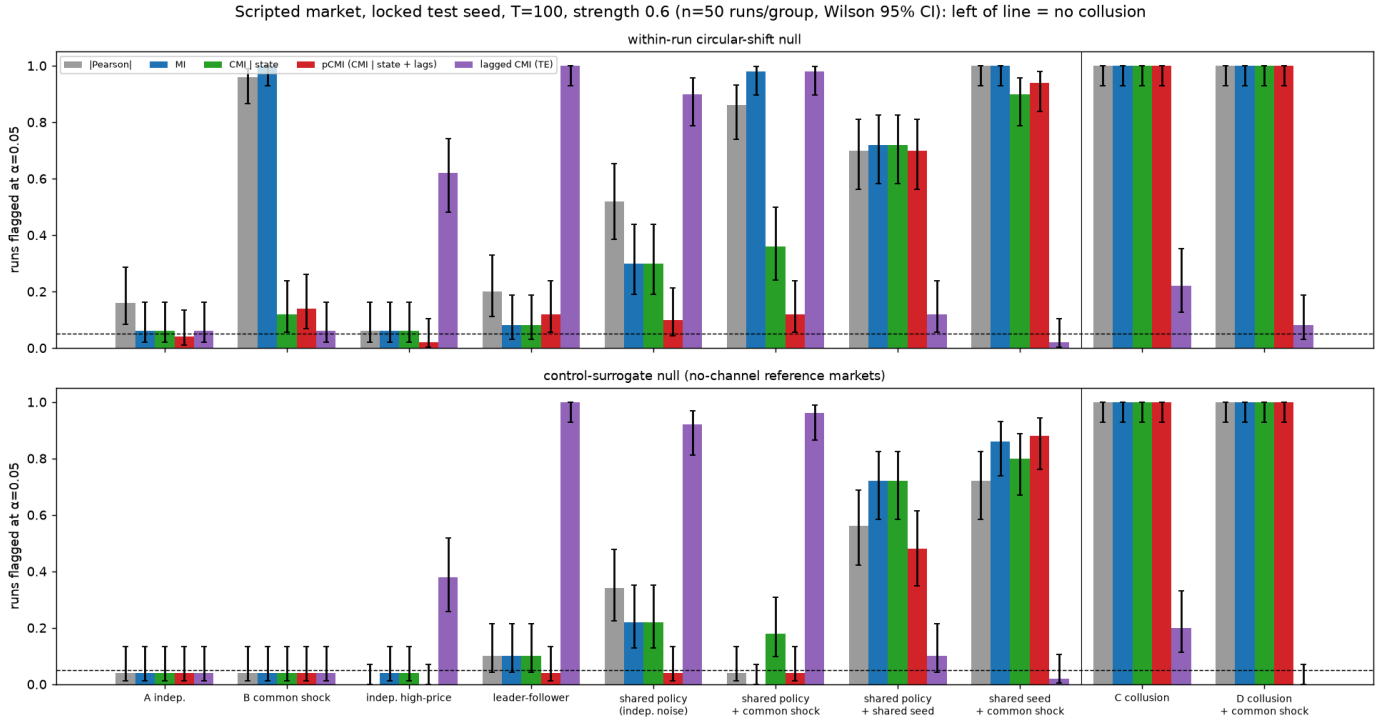


Figure 1 (Q1.4 §4.1, locked seed 1001, 50 runs per group). Run-level flag rates at α = 0.05.

Every contemporaneous detector flags and localises the planted pair in 100% of runs. Raw MI flags 100% of common-shock controls; conditioning on public state and lagged prices (pCMI) cuts this to 14% under the shift null and, for the held-out *shared policy + common shock* hard negative, 4% [1, 13] under the same-model surrogate null. Two failure modes are structural: two copies of one policy sharing a random seed are flagged by every contemporaneous detector (44–100%) — from price data they *are* a private shared signal — and lagged TE flags legitimate public leader–follower pricing in 100% of runs and shared policies in 90–98%, while finding the contemporaneous colluder in only 22%. Level screens fail elsewhere: unequal private costs are false-flagged in 30/30 scripted markets (pCMI 4/30; locked seed 2003).

5.2 With LLM sellers, the channel shows up as a price level; no co-movement is detected (Q1.4)

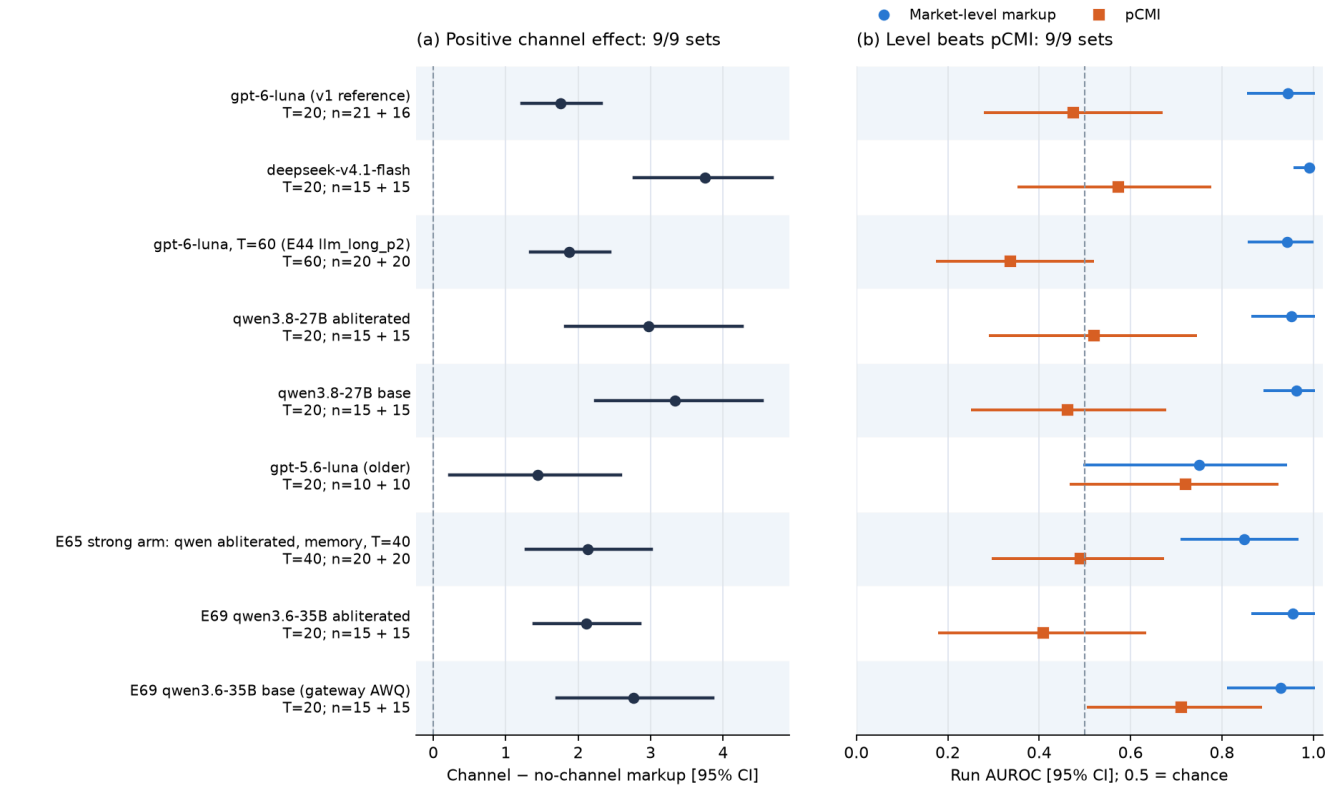


Figure 2 (Q1.4 P2.13). Nine main sets, seven seller models: channel − no-channel markup and run AUROC with game-bootstrap 95% CIs. Scores target channel assignment, not execution.

evidence	max-pair markup (level)	pCMI (co-movement)	source
9 main sets, 7 models	AUROC 0.75–0.99 (point est. > pCMI in 9/9); C − A CI > 0 in 9/9	0.34–0.72; CI ∋ 0.5 in 8/9	N1–N4
paired Δ level − pCMI	CI > 0 in 8/9 (exception gpt-5.6-luna +0.03 [−0.27, +0.33])		N3
E70 grid: 16-cell fractional factorial over 7 factors, 6 + 6 games per cell (192), board on	0.74 [0.66, 0.81]; level > pCMI in 12/16 cells (point est.)	0.51 [0.43, 0.59]	N8
E44 long games, T = 60 (pre-registered)	0.94 [0.86, 1.00]	0.34 [0.18, 0.52]	N5
executed channel games vs no-channel controls (48 executed of 149 labelled channel games, Q1.4 rule)	max-pair 0.99 (inflated by construction); pair-level 0.87 [0.80, 0.94] (not inflated)	0.42 [0.33, 0.53]	N7
rank (copula) pCMI / sign-change CMI, E70		0.53 [0.44, 0.61] / 0.44 [0.36, 0.52]	N11

Table 1. Per-set CIs uncorrected. E74's paired differences were declared, but declaration and results first appear in the same commit, so their order is not independently established.

Neither rank-based nor discrete estimators detect co-movement that linear pCMI missed (both at chance, though sign-CMI has little power in E70's sparse state strata). At T = 60 the observed co-move excess is +0.004 [−0.007, +0.015], whereas pCMI detects a planted shared ±1 move at per-round rate q = 0.1 with AUROC 0.79 [0.70, 0.88] — this is injection-specific sensitivity, not a general power bound. At T = 20 (most sets, all of E70) the same planting gives only 0.62 [0.52, 0.71], so small co-movement is not excluded there, and the check bounds sensitivity only to that planting. The one pCMI exception (base Qwen3.6, 0.71 [0.51, 0.88]) did not replicate in the grid's Qwen cells.

Lagged dependence is not excluded. TE is weakly above chance in the grid (0.62 [0.54, 0.70], declared secondary) and clearly above chance in some main sets (deepseek 0.81 [0.62, 0.96]; base Qwen3.8 0.91); level beats TE (CI > 0) in 6/9 main sets and in the grid (+0.12 [+0.01, +0.22]). Because TE also fires on

shared policies in scripted markets, and in the grid its signal sits in channel games that did *not* meet the execution rule (0.73, n = 52: 38 with a judged agreement, 14 without) rather than executed ones (0.49, n = 44; post hoc, difference untested), we do not read it as evidence of coordination.

Robustness to prompt wording is limited. Across five wordings (6 + 6 games each, gpt-6-luna), the channel effect C − A has CI > 0 in only 2/5, and no-channel markup ranges from −0.19 to +2.56 (P2.7). "9/9" holds for the main sets under one prompt family.

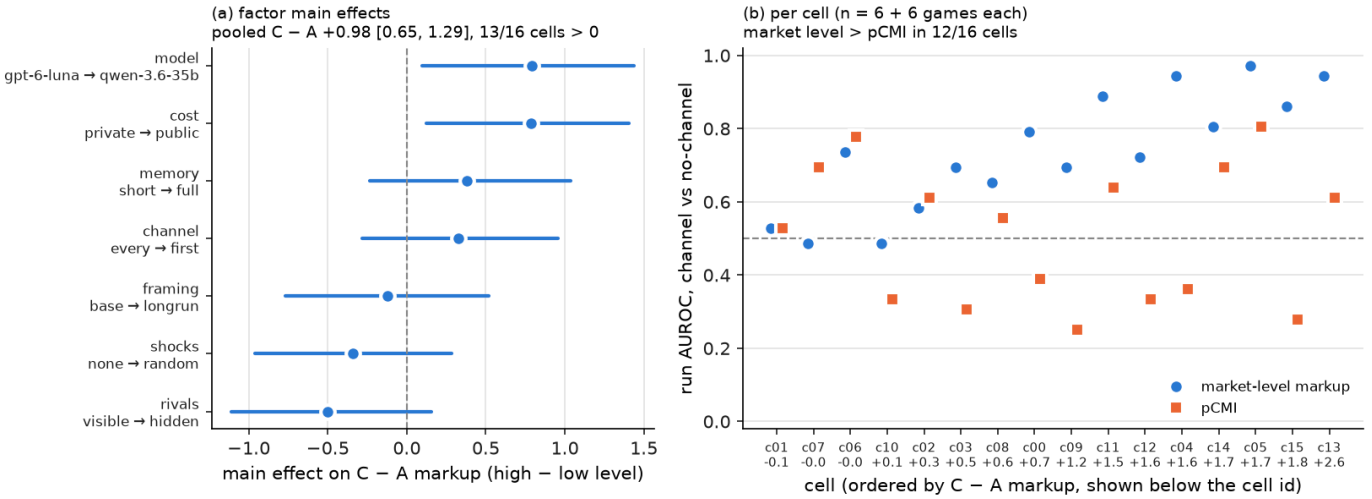


Figure 3 (Q1.4 P2.14). Registered 16-cell grid (E70), per-cell channel effect and detector AUROCs (cell estimates are descriptive).

5.3 A zero-shot auditor shows no resolved gain from public words — and can read them the wrong way (Q2.2)

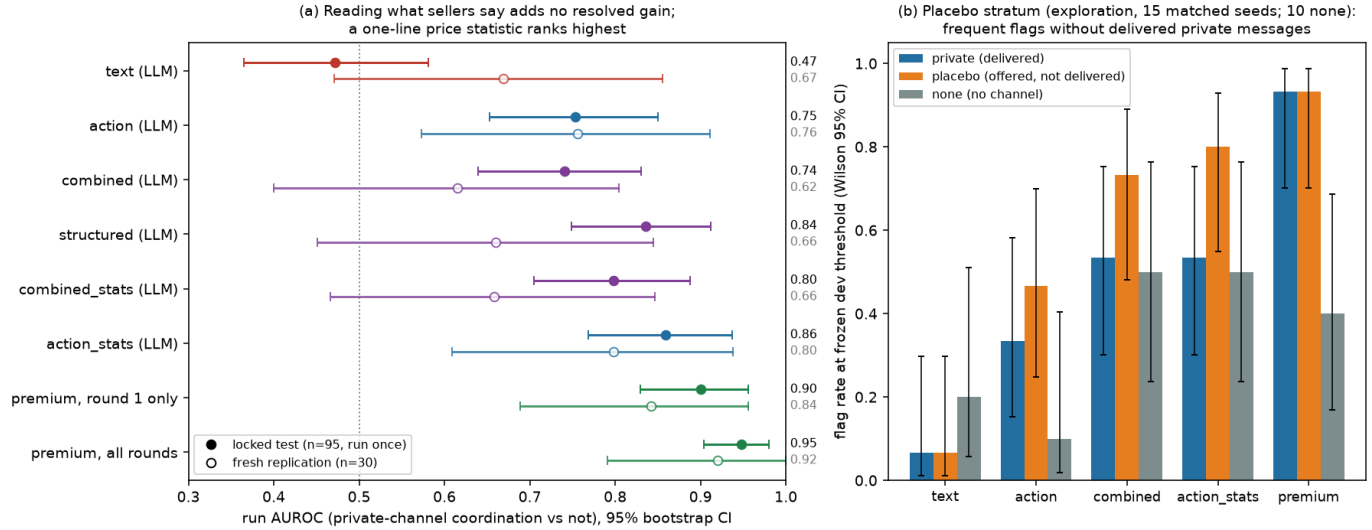


Figure 4 (Q2.2). (a) Run AUROC of each auditor view and of the one-line pair-premium statistic (chosen post hoc after v1), locked test (n = 95, run once) and the first fresh replication (`llm_v3rep`, n = 30; the pooled n = 90 contrast quoted in the text is not plotted). (b) Placebo stratum (15 matched seeds): flag rates when private messages are delivered vs offered but never delivered vs no channel. The x-axis label "private-channel coordination" means communication-positive vs negative on the test set and channel assignment vs controls on the replication, not verified coordination.

where text could help	contrast	n	text view	action view
luna, base framing (locked test)	judge-labelled y = 1 vs 0	95 runs	0.47 [0.36, 0.58]	0.75 [0.65, 0.85]
Claude Sonnet (talks publicly)	private vs none	5 + 5	0.22 [0.00, 0.52]	0.52 [0.20, 0.88]

where text could help	contrast	n	text view	action view
deepseek (talks publicly)	private vs none	30 runs	0.36 [0.20, 0.52]	0.53 [0.30, 0.74]
Qwen3.8 base	private vs none	15 + 15	**0.21 [0.12, 0.33]**	0.86 [0.69, 0.98]
Qwen3.8 abliterated	private vs none	15 + 15	**0.15 [0.05, 0.25]**	0.57 [0.36, 0.78]
SALT-NLP negotiation (external; act happens in chat)	rule-violating mutual ACCEPT vs not	240 episodes	0.87 [0.83, 0.91]	–

Table 2 (Q2.2 P2.0, `results/analysis/p2_say_vs_do.md`). AUROC < 0.5 = the view ranks controls higher; bold = CI entirely below 0.5.

The registered contrast shows no resolved gain from adding public text: combined − better single view −0.01 [−0.09, +0.06] (locked test, n = 95), −0.07 [−0.17, +0.02] on pooled fresh replications (n = 90, pooling post hoc); gains above +0.06 / +0.02 are excluded at 95%.

Inversion. An AUROC of 0.15 means the text carries information with the wrong sign for a zero-shot auditor, not that words are uninformative. Among the families in Table 2, the inversion is resolved only for the two Qwen3.8 arms (a gpt-6.1-sol seller arm with 0/7 private agreements is also inverted, 0.17 [0.01, 0.41], so inversion does not require an agreement). There, channel sellers, who can coordinate in private, invite rivals publicly *less* (invite rate 0.40 vs 0.88 in controls, Qwen3.8 base; post hoc); for deepseek and Claude invite rates do not differ and the mechanism is not established.

Cross-application to Q1.4's grid (E113, 192 games). In the 8 Qwen cells the LLM auditor reading prices is at chance (0.52 [0.41, 0.64]) — although Q1.4's markup statistic on the same games is not (0.76) — text is inverted (0.32 [0.24, 0.42]), and adding text lowers AUROC (combined − action −0.24 [−0.37, −0.09]; −0.30 [−0.44, −0.17] on games executed under Q1.4's rule). In the luna cells combined − action is +0.07 [−0.02, +0.17] overall and +0.16 [+0.05, +0.28] on executed games (registered secondary), but a similar +0.12 [−0.08, +0.32] appears in the 3 luna cells with no public posts (post hoc), so the luna gain cannot be credited to text. In these markets public text resolvedly hurt the auditor (Qwen) and nowhere resolvedly helped.

External case (different construct). In SALT-NLP bilateral negotiations, "executed" means a mutual ACCEPT that violates a verification rule — an act inside the conversation, not a pricing deal. There a text auditor reaches 0.87 [0.83, 0.91] (240 sampled episodes, 120 executed), even though 62% of executed episodes carry no explicit agreement (P(said | executed) = 0.385). It shows that text can work when the act is in the talk; it does not show that this is the only such condition.

Forecasting execution (E133, exploratory, n = 78 agreed games from Q1.4's grid, 48 held / 30 not under Q2.2's late-window rule). With privileged access to the private chat, say–do consistency forecasts which agreements are held (AUROC 0.77), but three hand-built say–do features in one logistic model add nothing beyond early price stability (0.855); richer text features are untested. An LLM reading the early public log *without* the private chat beats the best raw rule score (0.94 vs 0.885, Δ +0.06 [+0.02, +0.10], post hoc) — a compliance rule that uses early prices *and* the privately judged agreed price — with no evidence that its edge comes from public messages. This is the third analysis of the same 78 games; the rule score was fixed after E133c/d were seen, the LLM prompt states the label rule, and the same model (gpt-6.1-sol) is Q1.4's label judge and extracted the agreed price.

5.4 Neither signal is specific to collusion

- **Prompt artefact (Q2.2).** In a six-arm study (15 matched seeds), a private *notes-to-self* pricing notepad with no recipient raises the pair premium +1.04 [+0.65, +1.42] vs +1.61 [+0.92, +2.35] for the real channel (ratio 0.65 [0.34, 1.32]; the registered rule that the notepad is *not* the driver failed); an offered-but-never-delivered placebo gives +1.23 [+0.76, +1.85]; delivery itself (private − placebo, +0.38 [−0.41, +1.18]) is unresolved. Two paraphrases of the instructions (10 + 10 runs each) give +1.79 [+1.20, +2.43] and +0.24 [−0.32, +0.81]. Figure 5 shows the earlier placebo stratum: delivered and undelivered channels track each other from round 1. What the level screens detect is channel *assignment including its prompt cue*, not shown to be delivered coordination.

- **Tacit high prices (Q2.2; gpt-6-luna).** With memory and profit framing, 19/20 *no-channel* markets price ≥ Nash + 2; 13/20 channel pairs strict-execute, but the channel adds no detectable price, so channel labels stop being ground truth (Figure 6). This is model-specific: abliterated Qwen with memory (Q1.4 E65) is tacit-high in 0/20.
- **Prompt wording (Q1.4).** Wording alone moves no-channel markup from −0.19 to +2.56 (§5.2).
- **Legitimate asymmetry.** Scripted unequal costs: level screens false-flag 30/30 (Q1.4). LLM cost/quality edges: premium flags 7/10 and 9/10 vs controls 2/10 (Q2.2).
- **Prevalence.** At a hypothetical 1% prevalence none of the frozen, tested price flags reaches point-estimate precision 0.25 (Q1.4; maximum 0.20 [0.07, 0.54], which largely follows from α = 0.05; this is not a bound on achievable precision); the premium screen's point precision is 0.24 at 5% prevalence (Q2.2).
- **Labels disagree (Q2.2).** Only 14/50 communication-positive private test runs meet the E84 follow rule.

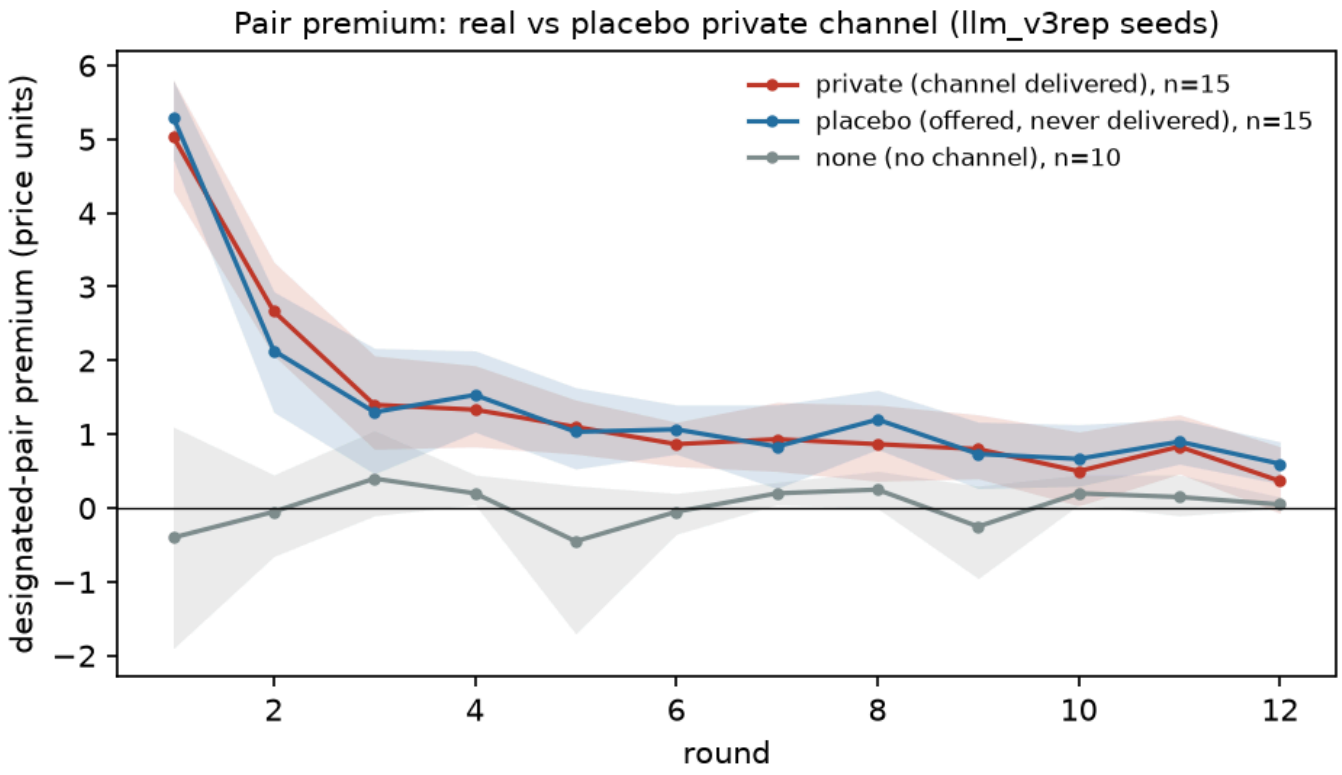


Figure 5 (Q2.2, `llm_v3rep` seeds). Designated-pair premium per round: delivered channel (n = 15) vs offered-but-never-delivered placebo (n = 15) vs no channel (n = 10). The premium is present before any message is delivered (round 1); placebo − private over rounds 2–12 is +0.03 [−0.54, +0.49] (no resolved difference, not equivalence).

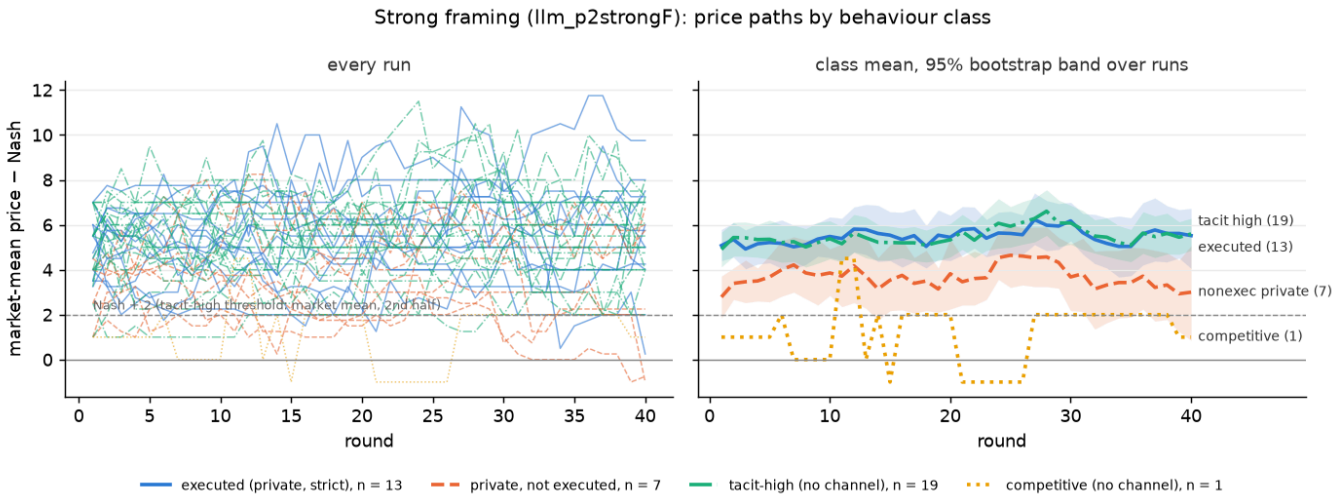


Figure 6 (Q2.2 P2.7, `llm_p2strongF`, T = 40). Memory + profit framing: market-mean price − Nash by behaviour class; 19 tacit-high no-channel markets sit with the 13 strict-executed channel pairs.

6. Discussion and implications

The two projects started from opposite bets — that collusion leaves a statistical fingerprint in actions, and that it leaves a semantic one in words — and converged: in these markets, channel assignment is visible mainly as **how high** prices are, not in detected **contemporaneous co-movement** (lagged TE is weakly above chance), and a zero-shot auditor shows no resolved gain from **what is said publicly** (small gains are not excluded). This is good news for cheap screening (a markup screen is simple and interpretable) and bad news for attribution: a high level has many innocent causes (wording, model, memory, legitimate cost advantages, market-wide tacit pricing), and private pricing deliberation can raise levels without delivered messages (Q2.2's logbook control suggests merely singling out a pair is insufficient, subject to instruction-length and batch-timing confounds).

Both projects state jointly that **detectability of LLM collusion by any price statistic is a property of the (seller model × prompt) pair**: relative screens (pair vs rivals) survive a shared model but lose power under market-wide spillover (in Q1.4's spillover sets pair_level 0.75–0.80 vs market level 0.85–0.94), and absolute screens need model- and prompt-matched references. With memory and a profit objective gpt-6-luna sellers raise prices market-wide without any channel (19/20; Q2.2 arm, re-scored by Q1.4 as E66), so channel labels stop being ground truth; abliterated Qwen with memory differs (channel adds +2.13 [1.28, 3.00]; 5/20 executed under Q1.4's rule; Q1.4 E65, T = 40). The joint statement is a post-hoc synthesis, not a tested prediction.

A practical reading is a two-stage design: an always-on level screen against a **model- and prompt-matched** reference as the first stage, with expensive audits (private-log review, dependence statistics, provenance checks for shared models and seeds) only on flagged pairs. We have not evaluated this pipeline end to end, and our evidence does not support recommending a particular architecture; it does support that a flag is an audit trigger, not a verdict.

7. Limitations

- **Ground truth.** Agreement labels come from one judge family (gpt-6.1-sol, also the auditor); the cross-family check is weak (above). Execution labels are defined from prices, so level screens are partly scored on their own input on executed subsets.
- **Scope of the co-movement null.** At T = 20 small co-movement is not excluded; sensitivity is bounded only for the planted ±1 strategy. Below-chance pCMI in some sets is unexplained.
- **Prompt dependence.** The channel effect is resolved in only 2/5 Q1.4 wordings; one Q2.2 paraphrase removes most of the premium.
- **No evasion test.** Monitor-aware sellers stopped agreeing (0/10, 0/10) rather than hiding; no red-teamed or instructed colluder was run.
- **Small samples, many comparisons.** 10–21 games per arm in most LLM sets; per-set CIs are uncorrected; E70's per-cell estimates are descriptive; several contrasts (pooled replication, E133e, E72, E113 luna split) are post hoc or secondary.
- **Narrow markets.** Four sellers, logit demand, short horizons; one external double auction (E68) is inconclusive (chat raises asks only +0.39 [+0.06, +0.69]); SALT is a different construct.
- **Placebo.** Q1.4's own market has no placebo arm; Q2.2's placebo leaves delivery unresolved (private − placebo +0.38 [−0.41, +1.18]).

8. Future work

1. Instructed / red-teamed colluders (coded public signals, gradual drift) to test evasion.
2. Ground truth independent of prices and of the condition, judged by more than one model family.
3. A larger placebo study with simultaneous pre-play to isolate delivered coordination.
4. Mixed-family markets and provenance data for the shared-model / shared-seed confound.
5. End-to-end evaluation of the two-stage audit against legitimate partnerships and tacit markets.

Reproducibility

Code, data and per-claim JSON live in the source repos (commits in SOURCES.md); each has a reproduce section (Q1.4 REPORT §6; Q2.2 REPORT Appendix E, `verify_phase2.py`: 31/31 commands re-run from cache with 0 LLM calls; it excludes sealed evaluations, E113/E115 data and SALT raw data). Figures in `figures/` are copies of the source figures. `SUBMISSION.pdf` and `SLIDES.pdf` are built by `uv run --group export python scripts/export_pdf.py`.

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Appendix — full tables

A. Q1.4 Phase 2 at a glance (verbatim from Q1.4 REPORT.md @ `0e06c03`; paths relative to Q1.4 `results/`)

[95% CI]: Wilson for k/n flag rates, Jeffreys for precision, class-stratified game bootstrap for AUROCs and markup differences. "Precision 0.25" in the shared row is a point-estimate threshold (maximum 0.20).

gap	question	key numbers	source (`results/…`)	§
shared	precision at realistic prevalence	at 1% prevalence no frozen flag reaches precision 0.25 (Jeffreys intervals for TPR/FPR)	`eval/prevalence.json`	P2.1
shared	independent reimplementation	422/424 run × statistic cells agree < 1e-9 (2 differ only in the ρ² cap); all 20 AUROCs identical; shift-null flags 4/21 vs 5/21 (Monte Carlo); no bug	`reimpl/compare.json`	P2.2
1	prompt wording (5 wordings, 6 + 6 each)	channel C − A +0.69 (SE 0.17), Q p = 0.69; no-channel baseline −0.19 … +2.56; within-wording markup AUROC 0.66 [0.53, 0.78]; pCMI 0.44 [0.30, 0.58]	`eval/p2_variants.json`	P2.7
2	long games (E44: T=60, 20 + 20, pre-registered)	**level, not short**: pCMI AUROC 0.34 [0.18, 0.52] at T=60 (0.52 at T=20); market level 0.94 [0.86, 1.00]; observed co-move excess +0.004 [−0.007, +0.015] < q = 0.1 planting (+0.13, pCMI 0.79); pair premium +0.70 → +0.23: rivals stay elevated while no-channel markets fall (spillover +2.24 of +2.47, rounds 51–60); executed 4/20, tacit 0/20	`eval/long_power_llm_long_p2_T60.json`, `eval/spillover_llm_long_p2_T60.json`	P2.8

gap	question	key numbers	source (results/…)	§
3	same-model confound	mismatched reference: pCMI 52% false flags at α = 0.10 (exploratory; fresh matched pool 3/21); model-matched fresh pool at α = 0.05: pCMI 2/21 [3, 29]% false flags, markup and pair_level 0/21 [0, 16]%, but markup power 1/16 (prompt mismatch)	`eval/same_model.json`, `eval/matched_ref.json`	P2.3
4	second seller family (deepseek, 15 + 15)	channel C − A +3.75 [2.77, 4.68] (luna +1.76; interaction +1.99 [0.90, 3.07]); AUROC markup 0.99, pair_level 1.00, pCMI 0.57 [0.36, 0.77]; rule HOLDS; judge: 15/15 agreements, 3/15 followed; public board alone (E59) +0.70 [0.23, 1.23], level screen false-flags 5/15	`eval/p2_family.json`	P2.9
5	coalition structure (scripted)	all-4 cartel: pair_level AUROC 0.26, 0.25 (A, B; 1/30, 0/30 flagged); markup, pCMI, TC 30/30	`p2/coalition_test.json`	P2.4
6	onset (scripted)	level CUSUM 0/30 false alarms, 30/30 detected, median delay 7–7.5 rounds; repeated-look pCMI 5/30 = 17% [7, 34] false	`p2/onset_test.json`	P2.4
7	market model (N = 2/4/6, μ)	pooled pCMI control false flags 18/300 = 6.0% [3.8, 9.3]; 9/10 cells pass; s = 0.6 power 30/30 everywhere	`p2/market_test.json`	P2.5
8	private cost asymmetry (hard negative)	pair_level and markup 30/30 false; pCMI 4/30 [5, 30]% (controls 1/30)	`p2/costasym_test.json`	P2.4
9	bid rotation (exploration)	predictable rotation: pCMI 6/30, 2/30 (controls 5/30, 4/30); TE 13/30, 17/30; levels 30/30	`p2/rotation_test.json`	P2.6
extra	sibling's six channel arms	undelivered notepad (0/14 agreements): pair_level names the pair 11/15, AUROC vs none 0.89 [0.76, 0.98], as private 12/15, 0.91	`eval/p2cue_ext.json`	P2.10
D2	external double auction (E68, qwen-3.6-35b, 15 + 15)	**inconclusive**: chat ask effect +0.39 [+0.06, +0.69] (≈2% of surplus); level AUROC 0.68 [0.46, 0.88], pCMI 0.60 [0.40, 0.80]; paid prices +2.61 vs +2.01; 18 parse failures in 14/30 runs	`eval/e68_cda.json`	P2.12
C	executed collusion, abliteration, model generations, memory (E61–E67)	executed 48/149 C games over ten sets (E44 T=60: 4/20 [0.07, 0.41]); tacit-high (no channel) 1/144 under our framing for current models, but 9/10 for older gpt-5.6-luna and 19/20 under Q2.2's strong framing (E66) (markup AUROC falls to 0.75); abliterated − base (C − A): no detectable increase in 3/3 Qwen pairs (E61 −0.37 [−2.09, 1.36], E69 −0.65 [−1.97, 0.65], Q2.2, pair premium not C − A, −1.24 [−2.41, −0.03]); executed-only pooled pCMI 0.42 [0.33, 0.53] vs 0.52 non-executed (market_level on that subset is inflated by construction, 0.99; pair_level is not, 0.87 [0.80, 0.94]); E69 base Qwen3.6 is the one set with pCMI CI > 0.5 (0.71 [0.51, 0.88]; market level 0.93); E65 memory arm (T=40): markup 0.85, pCMI 0.49, declared executed-only pCMI test inconclusive (0.33 [0.02, 0.66], 5/20 executed; the pilot's 5/5 did not replicate); E66 (Q2.2 strong framing, gpt-6-luna, T=40): executed 13/20 but tacit 19/20 without a channel, so channel vs no-channel is 0.46 on pCMI and 0.44 on market level (nothing separates; declared test not evaluable); E67 stickiness screen: 1 of 3 confirmatory sets (E65 0.90; gpt-5.6-luna 0.46, E66 0.42), so the stickiness explanation of low pCMI is not supported; glm-5.2 dropped	`eval/p2_partc.json`	P2.11
E1	generality over 7 design factors (E70, 2^(7−3) grid, 16 cells × 6 + 6, registered)	channel C − A positive in 13 / 0 / 3 cells (pos / tie / neg; one-sided sign p = 0.011), pooled +0.98 [0.65, 1.29]; level > pCMI 12 / 1 / 3; pooled AUROC level 0.74 [0.66, 0.81], pCMI 0.51 [0.43, 0.59], TE 0.62 [0.54, 0.70] (declared secondary: confirmed); qwen pCMI exception not replicated 0.46 [0.35, 0.58]; largest main effects M (= model incl. ≈100% vs 4–10% public-board use) +0.79 [0.10, 1.44], cost +0.79 [0.12, 1.41] (descriptive); follow-up on M not confirmed: +1.07 [−0.20, +2.41] (12 + 12 per level)	`eval/e70_grid.json`	P2.14

B. Q2.2 claims × denominators (verbatim from Q2.2 `results/analysis/claims_denominators.md` @ `8b680ee`, generated by `p2_claims.py`)

id	claim	value (abstract text)	status	denominator / registration	source : key path
C01	combined minus better single view, run-AUROC, locked test	-0.01 [-0.09, +0.06]	OK	n=95 runs (llm_v3 locked test split of 125 runs); registered primary contrast	`results/analysis/final_llm_v3_gpt-6.1-sol_test.json : contrasts['delta_hybrid[combined]:run_auroc'] {delta,ci}; n_runs`

id	claim	value (abstract text)	status	denominator / registration	source : key path
C02	same contrast, two fresh replications pooled	-0.07 [-0.17, +0.02]	OK	n=90 runs (30 llm_v3rep + 60 llm_p2rep, label = channel condition); post hoc pooling of two registered splits	`results/analysis/p2_pool_rep.json : condition_label.contrasts['delta_hybrid[combined]'] {delta,ci}; n; n_by_ds`
C03	gains above +0.06 (test) / +0.02 (pooled) excluded at 95% (upper CI limits)	+0.06 / +0.02	OK	upper 95% CI limit of C01 (n=95) and C02 (n=90); pooled is post hoc	`results/analysis/final_llm_v3_gpt-6.1-sol_test.json + results/analysis/p2_pool_rep.json : contrasts[...run_auroc].ci[1]; condition_label.contrasts['delta_hybrid[combined]'].ci[1]`
C04	a computed pair-statistics table added to the action view gains on test (action_stats minus action)	+0.11 [+0.04, +0.18]	OK	n=95 runs, locked test; registered contrast family (delta_hybrid[action_stats] = action_stats - max(text, action)), not the primary endpoint. Corrected 2026-10-24 16:40: the abstract previously quoted structured (+0.08)	`results/analysis/final_llm_v3_gpt-6.1-sol_test.json : contrasts['action_stats-action:run_auroc'] {delta,ci}`
C04b	adding text to the pair-statistics view does not gain (combined_stats minus action_stats)	-0.06 [-0.14, +0.01]	OK	n=95 runs, locked test; registered contrast family	`results/analysis/final_llm_v3_gpt-6.1-sol_test.json : contrasts['combined_stats-action_stats:run_auroc']{delta,ci}`
C04c	same pair-statistics gain on fresh data (action_stats minus better single view), two replications pooled	+0.00 [-0.06, +0.07]	OK	n=90 runs (30 llm_v3rep + 60 llm_p2rep, label = channel condition); post hoc pooling; added 2026-10-24 17:57 (iter26 review: test gain did not replicate)	`results/analysis/p2_pool_rep.json : condition_label.contrasts['delta_hybrid[action_stats]'] {delta,ci}`
C05	pair price premium AUROC, fresh replication	0.94 [0.87, 0.99]	OK	n=60 runs (llm_p2rep: 30 private / 20 none / 10 legit); premium selected post hoc after v1; replication registered	`results/analysis/replication_llm_p2rep.json : point_ci.premium.run_auroc; n_runs; group_n`
C06	unread pricing notepad premium as share of channel premium (rounds 2-12)	65% (1.04/1.61)	OK	n=15 seeds per arm (llm_p2cue: notes vs none, private vs none, paired by seed); gpt-6-luna; registered arms, ratio is derived	`results/analysis/p2_cue_llm_p2cue.json : paired['notes-none'].prem_r2_12[0] / paired['private-none'].prem_r2_12[0]; n`
C07	legitimate cost / quality edges flagged by premium vs controls	7/10, 9/10 vs 2/10	OK	10 runs per condition (highcost, quality, plus 10 no-channel reference runs from llm_p2cue); gpt-6-luna; frozen dev threshold 0.667	`results/analysis/p2_asym.json : conditions.{highcost,quality,none_ref_p2cue}.premium_flags`
C08	text views flag every control (no-channel) market for Claude, deepseek, Qwen base+abliterated	100% (5/5, 15/15, 15/15, 15/15)	OK	control (condition=none) markets: Claude 5, deepseek 15, Qwen-base 15, Qwen-abliterated 15; text view; families differ by model so cross-family comparison is associative; registered predictions (E95) for Claude/deepseek	`results/analysis/p2_family3way.json + results/analysis/p2_strong_eval_llm_p2qwena.json + results/analysis/p2_strong_eval_llm_p2qwenb.json : llm_p2{claude,ds}.views.text.flags_by_class.PUB; llm_p2qwen{a,b}.detectors.text.flags.none`

id	claim	value (abstract text)	status	denominator / registration	source : key path
C09	text-view run AUROC range across those control-flagging families (private vs none)	0.15-0.36	OK	run AUROC private vs none, text view: Claude n=10, deepseek n=30, Qwen-base n=30, Qwen-abl n=30 runs; TL;DR claim (not in abstract text)	`results/analysis/p2_family3way.json + results/analysis/p2_strong_eval_llm_p2qwen{a,b}.json : views.text.auroc_private_vs_none[0] (min,max over 4)`
C10	SALT-NLP episodes	8000	OK	episodes = 7998 with valid judge label + 2 invalid-judge episodes; 800 runs; external data (SALT-NLP)	`results/analysis/salt_d1.json : E104_sayvsdo.n_episodes + n_invalid_judge; crosstab sum`
C11	SALT distinct models	10	OK	distinct model names parsed from per_condition ids (main__X and cross_model__A__B); 16 conditions x 500 episodes	`results/analysis/salt_d1.json : E104_sayvsdo.per_condition[*].condition (parsed)`
C12	share of executed SALT episodes with no explicit agreement (1 - P(said|exec))	62% (P(said|exec)=0.385)	OK	executed episodes n=5169 (1989 said + 3180 not said), 7998 judged; Qwen judge label for 'said'; episodes are clustered in 800 runs (check CI method in salt script)	`results/analysis/salt_d1.json : E104_sayvsdo.P_said_given_exec.est; crosstab.exec1_*`
C13	test 'colluders' that keep their agreement	14/50	OK	n=50 judged-agreement private runs among the 95 llm_v3 test runs (14 followed + 36 not followed); behaviour labels registered (E84)	`results/analysis/p2_behaviour.json : datasets.llm_v3.class_counts.{followed,not_followed}; n`
C14	strict-executed channel pairs (memory + profit objective)	13/20	OK	20 channel markets (llm_p2strongF, seeds 9800000-9800019); strict rule fixed in registered amendment E99-amend; Jeffreys [0.43, 0.83]	`results/analysis/p2_strong_classes_llm_p2strongF.json : n_strict_executed`
C15	no-channel markets that price >= Nash + 2 (tacit-high)	19/20	OK	20 no-channel markets (paired seeds with C14); tacit-high rate >= 0.5 registered (E99amend); Jeffreys [0.79, 0.99]	`results/analysis/p2_strong_classes_llm_p2strongF.json : n_tacit_high_none`
C16	paired excess private - none (market price over cost-adjusted Nash)	-0.44 [-1.84, +1.00]	OK	n=20 paired seeds; registered prediction E99_P3 (excess > 0) FAILED; unresolved	`results/analysis/p2_strongF_report_llm_p2strongF.json : E99_P3.paired_excess_private_minus_none; n_pairs`
C17	private-plan LLM monitor AUROC, strict-executed vs tacit-high	0.76 [0.55, 0.93]	OK	13 strict-executed vs 19 tacit-high markets; EXPLORATORY (E103 plan view registered before plan calls, but classes defined post hoc); CI [0.55, 0.93]	`results/analysis/p2_strong_eval_llm_p2strongF.json : detectors.plan.auroc_strict_vs_tacit_high (== keyword file detectors.plan_view...); groups_n`
C18	no-LLM count of plans naming the chat partner, same contrast	0.82 [0.66, 0.96]	OK	13 vs 19 markets; EXPLORATORY post hoc keyword detector (E111) K_dir; CI [0.66, 0.96]	`results/analysis/p2_plan_keyword_llm_p2strongF.json : detectors.K_dir.auroc_strict_vs_tacit_high; n_pos_strict; n_neg_tacit_high`
C19	no plan uses agreement words	0/13 strict, 0/19 tacit-high	OK	13 strict + 19 tacit-high (+7 non-executed) markets; keyword K_terms; exploratory	`results/analysis/p2_plan_keyword_llm_p2strongF.json : runs_with_term_in_any_plan`
C20	public-text view AUROC, strict-executed vs tacit-high (inverted)	0.25 [0.08, 0.46]	OK	13 vs 19 markets; EXPLORATORY (E99-x, added after generation began, before detector scores); CI [0.08, 0.46]	`results/analysis/p2_strong_eval_llm_p2strongF.json : detectors.text.auroc_strict_vs_tacit_high`

id	claim	value (abstract text)	status	denominator / registration	source : key path
C21	abliterated vs base Qwen: executed markets each arm; agreement at ceiling	13/15; 13/15	OK	15 vs 15 private markets (llm_p2qwena vs llm_p2qwenb); execution diff 0.00 [-0.26, +0.26] Newcombe; agreement 15/15 both; registered (C3), predictions not supported	results/analysis/p2_qwen_cmp.json : P2_executed_rate.abliterated; P2_executed_rate.base; P1_agreement_rate
C22	SALT-NLP: gpt-6.1-sol text auditor AUROC for execution	0.87 [0.83, 0.91]	OK	n=240 sampled SALT episodes (120 executed; 79 tacit) audited by gpt-6.1-sol (E105); external data	results/analysis/salt_d1.json : E105_audit.auroc{est,ci}; n; n_exec
C21b	abliterated vs base Qwen: Q2.2 strict-executed each arm (lower-bound compliance)	5/15; 5/15	OK	15 private markets per model; Q2.2 strict: F>=12, both pair prices>=F-1 in >=50% of later rounds, >=10 later rounds, post-agreement pair excess>=2; p2_strong.py via p2_strong_eval.py, not a +/-1 band or Q1.4 E70 rule	results/analysis/p2_strong_classes_llm_p2qwena.json : n_strict_executed (qwena; qwenb)
C23	abliterated minus base Qwen: paired channel premium difference	-1.24 [-2.41, -0.03]	OK	15 vs 15 private markets with matched none; bootstrap; registered P3 (abliterated >= base) not supported; bf16 vs FP8 serving confound	results/analysis/p2_qwen_cmp.json : P3_paired_premium.diff_boot95
C24	E116 within channel: plan view AUROC strict-executed vs non-executed	0.69 [0.44, 0.91]	OK	13 strict vs 7 non-executed private markets (llm_p2strongF); exploratory, zero LLM calls, seed bootstrap	results/analysis/p2_within_agree_llm_p2strongF.json : views.plan.auroc_strict_vs_nonexec
C27	E116b within agreement, Qwen abliterated / base: public text AUROC strict vs non-executed	0.73 [0.43, 0.99]; 0.51 [0.25, 0.70]	OK	5 strict vs 10 non-executed private markets per model (llm_p2qwena, llm_p2qwenb; 15/15 judged agreement each); exploratory, not pooled	results/analysis/p2_within_agree_llm_p2qwen{a,b}.json : views.text.auroc_strict_vs_nonexec
C25	E112 participant replacement (luna newcomer), private arm: market excess replaced - original; re-agreement replaced; originals	+0.20 [-0.08, +0.62]; 16/20; 15/20	OK	20 paired private games replayed from llm_p2strongF, newcomer from round 21, rounds 21-40; registered E112, all 4 predictions held; none arm +0.16 [-0.27, +0.57] in P2.15	results/analysis/p2_replace.json : arms.private.01_market_repl_minus_orig; arms.private.04_reagreement_floor_ge13.{repl,orig}
C28	E115 participant replacement (gpt-6.1-sol newcomer), private arm: re-agreement replaced; markets >= Nash + 2 replaced	0/20; 18/20	OK	same 20 paired private games as E112, sol newcomer from round 21; registered E115, S1 failed, S2-S4 held; none arm 9/10 in P2.15	results/analysis/p2_replace_sol.json : arms.private.04_reagreement_floor_ge13.repl; arms.private.06_market_excess_ge2.repl
C29	E120 definition sensitivity (P2.7 body): strict-executed one-at-a-time range; full grid; tacit-high range	10-16/20; 7-18/20; 17-20/20	OK	llm_p2strongF, frozen judge terms, 81 threshold combinations; exploratory, zero LLM calls; registered 13/20 reproduced	results/analysis/p2_strict_sensitivity.json : strict_range_one_at_a_time; strict_range_all; tacit_high[].k
C30	E112/E115 post-replacement executed counts under registered replacement O5 (not Q2.2 strict)	15/20; 14/20; 0/20; 14/20	OK	20 paired private games per replacement arm; O5: judged post-r20 agreement F>=13, both pair prices>=F-1 in >=50% of later rounds, >=5 later rounds, post-agreement pair excess>=2; same rule for originals; REPORT P2.7/P2.15	results/analysis/p2_replace.json; results/analysis/p2_replace_sol.json : arms.private.05_executed_after_r21.{repl,orig}
C31	E113 Qwen cell-stratified combined minus action / text AUROC	-0.24 [-0.37, -0.09]; -0.05 [-0.13, +0.03]	OK	8 Qwen cells x 6 matched seeds x 2 arms = 96 games of Q1.4 E70's 192; channel assignment label; equal-cell mean of within-cell AUROC differences; game bootstrap within cell x arm, B=2000 seed=0; E113-amend3 registered before scores; shared Q1.4 data, not independent joint replication	results/analysis/p2_e1_audit.json : amend3.qwen_stratified_combined_minus_{action,text}.{delta,ci,n_cells}

id	claim	value (abstract text)	status	denominator / registration	source : key path
C32	E113 combined-minus-best-single cell signs: Qwen positive/tie/negative; all cells positive/tie/negative	0/8; 0; 8; 4/16; 1; 11	OK	8 Qwen cells and all 16 Q1.4 E70 cells, 6 games per arm per cell; best-single max selection biases delta downward; ties count toward registered P1, so held P1 is weak confirmation	`results/analysis/p2_e1_audit.json` : `amend3.{qwen,all}_sign_counts; b_cells_combined_gt_best_single; b_P1_held`
C33	E113 Qwen text inversion and flags in channel / no-channel arms	0.32 [0.24, 0.42]; 48/48; 48/48	OK	96 games in 8 Qwen E70 cells (48 channel, 48 no-channel); unstratified pooled text AUROC and frozen dev threshold flags; game-bootstrap CI, not an independent replication of Q1.4 data	`results/analysis/p2_e1_audit.json` : `pooled_auroc_channel_vs_none_descriptive.text_qwen; a_text_flag_rate_by_arm_qwen.{1,0}`
C34	E133e no-chat LLM minus best raw early-price rule score; oriented say-do gap and pair SD AUROC	+0.06 [+0.02, +0.10]; 0.77; 0.855	OK	78 Q1.4 E70 grid agreed games (48 held / 30 not); post hoc robustness declared before computing (E133e-reg); paired game bootstrap; 0 LLM calls	`results/analysis/e133e_attr.json` : `b_llm_E133d_minus_raw.rule_score; a_oriented_auroc_smaller_is_held.{say_do_gap,pair_sd_0_5}[0]`